# Motors and generators

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Distinguish electrical-to-mechanical transfer from mechanical-to-electrical transfer.

**SUCCESS CRITERIA - I can...**

1. Trace energy through a motor and a generator.
2. Identify magnetic interactions in rotation or induction.
3. Include less useful energy transfers.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A consistently drawn magnetic-field map has **3** times as many field lines through the same small perpendicular area at A as at B. What does the denser spacing indicate?

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $\eta = P_{useful}/P_{input}$; $E_{input} = E_{useful} + E_{losses}$. Motor: electrical to mechanical; generator: mechanical to electrical. Count losses and any stored-energy change.

**Worked reasoning:** An idealized motor receives **20 J** electrically and delivers **12 J** mechanically; **8 J** goes to other transfers. A generator runs the energy conversion in the opposite useful direction.

**Your turn:** Draw separate motor and generator energy-flow diagrams.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

**Complete two energy chains**: battery → motor → turning axle, and turning axle → generator → lamp. **Label each input** and useful output.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A small classroom car motor receives **100 J** of electrical energy and delivers **60 J** of useful mechanical energy. **Complete the energy-flow diagram**, calculate efficiency and **name the energy transferred** to other forms.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

**Marzano check - circle:** ① ② ③

## 05 Explain your reasoning

A motor drives a generator that feeds the motor. **Explain why real losses prevent** the loop from running forever without external input.

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

An electric field of **6 N/C** points left. **Draw the force direction** on **+4 C** and on **-4 C**, then **compare sizes**. **Draw field** and force arrows separately for each charge.

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

**Marzano check - circle:** 1 2 3

## 07 Exit check

Which device converts mechanical input into electrical output?

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

A motor draws the same electrical power in two designs, but one delivers more mechanical power. **Compare efficiencies** and **name a measurement needed**.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

**Marzano check - circle:** 1 2 3